

ADDIS ABABA BOLE INTL, Ethiopia

WMO: 634500

Lat: 8.978N

Lon: 38.799E

Elev: 7630

StdP: 11.07

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	45.5	48.0	19.4	19.4	62.9	24.8	25.1	61.1	17.3	61.3	14.9	63.2	4.2	330	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	20.2	79.1	56.9	77.9	57.0	76.8	57.1	64.3	72.6	62.7	71.3	61.3	70.2	12.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	107.7	70.2	59.4	100.8	67.5	58.3	96.9	65.9	33.9	72.9	32.4	71.0	31.4	70.9	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.4	16.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	61.8	64.2	66.2	65.7	65.9	64.4	61.9	61.4	62.2	62.8	62.1	60.7
	DBStd	2.94	2.48	2.51	2.72	2.48	2.43	2.40	1.91	1.69	1.91	2.06	2.22	2.34
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	835	102	41	18	21	19	39	99	113	86	72	90	135
	CDD50	4833	365	396	502	472	492	431	367	352	366	397	362	331
	CDD65	196	3	17	55	44	46	21	1	0	2	4	2	1
	CDH74	1181	57	162	328	220	252	116	2	0	4	19	13	8
	CDH80	22	1	1	10	5	4	1	0	0	0	0	0	0
(14) Wind	WSAvg	7.8	8.4	9.3	8.7	7.8	8.0	6.8	6.5	6.0	6.1	9.0	8.8	8.8
(15) Precipitation	PrecAvg	47.1	0.7	1.4	2.6	3.4	3.2	5.0	10.0	11.0	6.7	1.5	0.4	0.4
	PrecMax	61.3	5.0	6.6	9.2	9.0	12.8	10.2	15.6	16.8	15.2	10.7	2.4	2.6
	PrecMin	35.6	0.0	0.0	0.0	0.2	0.0	1.5	7.1	4.9	2.2	0.0	0.0	0.0
	PrecStd	6.1	0.9	1.5	2.1	2.0	2.5	1.8	1.8	2.6	2.4	1.9	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	79.1	81.0	80.6	80.4	79.1	73.4	72.0	74.5	75.6	75.4	75.3
		MCWB	54.9	55.6	56.3	57.6	57.1	57.4	60.7	60.8	58.7	57.4	57.8	55.7
	2%	DB	75.3	77.3	79.1	78.4	78.6	77.0	70.9	70.0	72.1	74.2	73.7	73.5
		MCWB	54.5	56.0	56.2	57.7	57.8	58.1	59.5	60.2	59.0	57.0	57.4	55.6
	5%	DB	73.7	76.0	77.3	76.6	77.0	75.1	69.2	68.2	70.5	73.1	72.5	72.0
		MCWB	54.1	55.7	56.2	58.2	57.7	58.2	59.3	59.4	58.9	56.6	57.0	54.9
	10%	DB	72.0	74.6	75.4	74.8	75.1	72.9	67.6	66.5	68.9	71.6	71.3	70.9
		MCWB	53.8	55.4	55.9	58.2	57.9	58.8	58.9	58.8	58.8	55.6	56.5	54.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.7	65.4	66.1	65.0	66.0	64.4	62.7	63.0	63.2	62.6	62.8	61.8
		MCDB	72.9	74.6	75.0	73.6	75.4	72.1	69.3	69.0	69.9	71.9	70.9	71.3
	2%	WB	59.4	61.8	61.8	62.5	63.5	62.2	61.0	61.1	61.2	60.2	60.5	59.1
		MCDB	71.4	72.9	71.1	71.7	72.8	71.3	68.4	67.5	68.5	70.4	70.1	70.2
	5%	WB	58.0	59.2	59.6	60.9	61.3	60.7	60.0	60.1	60.2	58.7	59.1	57.5
		MCDB	69.9	70.6	70.8	71.2	71.0	70.1	67.1	66.4	67.6	69.0	69.5	68.9
	10%	WB	56.7	57.6	58.3	59.8	59.9	59.6	59.1	59.2	59.3	57.7	57.9	56.1
		MCDB	68.2	69.2	69.7	70.2	69.8	69.0	65.8	65.3	66.6	67.7	68.1	67.6
(35) Mean Daily Temperature Range	5% DB	MDBR	23.2	22.8	20.2	18.7	18.4	17.6	13.0	12.8	14.8	19.0	20.3	22.5
		MCDBR	25.3	24.2	22.0	21.9	21.8	21.2	15.3	14.8	17.0	21.1	21.7	23.7
		MCWBR	12.2	11.3	9.1	9.2	9.8	9.8	7.9	8.0	9.0	10.8	11.8	11.9
	5% WB	MCDBR	21.2	21.3	19.4	17.7	18.2	18.1	14.2	13.8	15.1	18.2	19.3	20.9
		MCWBR	12.5	12.1	9.3	9.7	10.5	10.1	8.0	8.1	9.0	10.6	11.9	12.4
(40) Clear-Sky Solar Irradiance	taub		0.312	0.333	0.361	0.368	0.405	0.505	0.497	0.432	0.412	0.337	0.306	0.301
	taud		2.513	2.461	2.407	2.419	2.268	1.934	1.981	2.208	2.248	2.456	2.532	2.537
	Ebn at Noon		318	313	304	297	281	250	253	275	284	308	318	319
	Edh at Noon		34	36	39	38	43	60	57	47	45	36	33	32
(44) All-Sky Solar Radiation	RadAvg		1945	2102	2093	1979	1964	1827	1600	1564	1748	1968	1937	1908
	RadStd		130	145	167	120	103	72	118	104	88	104	80	163

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page